

Demonstrating FoolProofJoint: Reducing Assembly Errors of Laser Cut 3D Models by Means of Custom Joint Patterns

Keunwoo Park
Hasso Plattner Institute, University of
Potsdam, Germany
keunwoo.park@guest.hpi.de

Conrad Lempert
Hasso Plattner Institute, University of
Potsdam, Germany
conrad.lempert@hpi.de

Muhammad Abdullah
Hasso Plattner Institute, University of
Potsdam, Germany
muhammad.abdullah@hpi.de

Shohei Katakura
Hasso Plattner Institute, University of
Potsdam, Germany
shohei.katakura@hpi.de

Jotaro Shigeyama
Hasso Plattner Institute, University of
Potsdam, Germany
jotaro.shigeyama@hpi.de

Thijs Roumen
Hasso Plattner Institute, University of
Potsdam, Germany
thijs.roumen@hpi.de

Patrick Baudisch
Hasso Plattner Institute, University of
Potsdam, Germany
patrick.baudisch@hpi.de

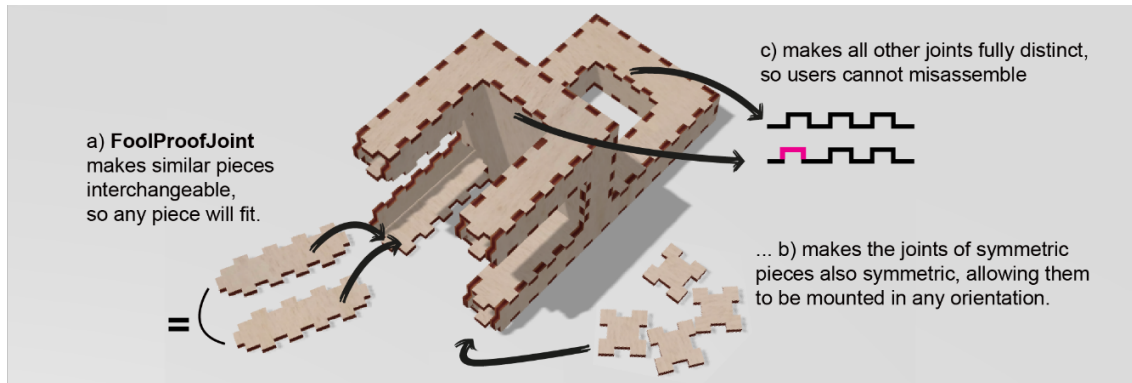


Figure 1: To eliminate assembly errors, FoolProofJoint optimizes finger joint patterns, to (a) make pieces interchangeable by giving them the same joint patterns where pieces have an identical shape, (b) prevent symmetric pieces from being mounted in the wrong orientation, and (c) prevent incorrect pieces from being assembled.

ABSTRACT

We demonstrate FoolProofJoint, a software tool that simplifies the assembly of laser-cut 3D models and reduces the risk of erroneous assembly. FoolProofJoint achieves this by modifying finger joint patterns. Wherever possible, FoolProofJoint makes similar looking pieces fully interchangeable, thereby speeding up the user's visual search for a matching piece. When that is not possible, FoolProofJoint gives finger joints a unique pattern of individual finger placements so as to fit only with the correct piece, thereby preventing erroneous assembly.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

KEYWORDS

Personal fabrication, laser cutting, rapid prototyping, manual assembly

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1 INTRODUCTION

In recent years, laser cutting has become one of the main contenders for fast fabrication. The cutting process itself is fast, it is a subtractive fabrication method which can cut an entire model within minutes [2]. Furthermore, recently developed specialized 3D modeling software, such as FlatFitFab [6], Platener [4], and Kyub [3] allow generating the required 2D cutting plans quickly, while allowing users to model efficiently in 3D.

With fast 3D design and fast fabrication in place, assembly has become the bottleneck that hinders further speed-ups [1]. Assembly requires users to perform a sequence of steps, each of which typically involves the user connecting a piece with another piece based on a pair of matching joints.

Several systems offer methods to facilitate correct assembly, typically with the help of some visual language that communicates which joint to connect to which other joint. FlatFitFab [6] and Kyub [3], for example, engrave matching pairs of numbers; Roadkill [1] instead “points” users to the matching piece directly from within the 2D layout; Daedalus in the dark [5] laser cuts haptic paths in the plates to make such assembly instructions are accessible to blind users.

But assembly instructions have limitations. First, since engraving assembly instructions into the model arguably ruins the appearance of that model, more recent systems cut or engrave next to the pieces [1]. Once a piece is removed from the layout though, piece and instructions are no longer associated, allowing things to go wrong. Second, even with instructions engraved into the pieces, we have observed participants of various laser cutting workshops getting things wrong from time to time, simply due to human error.

We demonstrate FoolProofJoint [7], a software tool that minimizes visual search during assembly, by making similar pieces fully interchangeable (Figure 1a and b) and prevents assembly errors by making finger joints distinct (Figure 1c). FoolProofJoint is integrated into Kyub [3] and modifies joints in real-time, as the user is modeling. We point readers to the full paper [7] for implementation and evaluation details.

2 FOOLPROOFJOINT

When users assemble a 3D laser-cut model (Figure 2), they find themselves looking at a 2D layout of pieces. Users pick up a piece and then look for the matching piece to attach. They limit their search to pieces that feature an edge of matching length. In addition, they may have expectations about the overall shape of the desired piece that help them select which pieces to assemble.

Two types of assembly errors tend to emerge. First, users may reach for an incorrect piece. Second, for pieces with a symmetric envelope, users may try to assemble a piece in the wrong orientation. In the following, we go over these two classes of assembly errors and how FoolProofJoint addresses them.

2.1 Issue 1: Users pick up similar looking pieces

Figure 3a illustrates what happens during assembly. Here the user is trying to assemble one of the chair’s legs. The user is looking for a piece that could fit, thus considers all pieces that feature roughly the dimensions of a leg.

Ideally, the user picks the correct piece and successfully assembles it. However, this may not be the most likely outcome, given that there are eight pieces of the same envelope and two more pieces with a similar envelope. There are three other possible outcomes with increasingly dire consequences.

(1) The user picks up an incorrect piece, tries to assemble it, and recognizes that it does not fit because their joints collide. This case causes little damage, as the user simply drops the piece and tries a different one.

(2) Less ideally, the incorrect piece still fits. The user attaches it, to then realize that it is incorrect, e.g., because adjacent joints are not aligned. The user removes the piece again and tries a different one. Removing a piece, however, produces a lasting negative effect on both pieces involved, as it wears out their joints, making the final model less sturdy.

(3) The worst possible outcome is illustrated in Figure 3a: an erroneous piece may fit, and the error may go unnoticed, as subsequent pieces continue to fit as well, thereby producing a propagation error. (b) By the time the user recognizes the issue, the user has to dis-assemble *multiple* pieces.

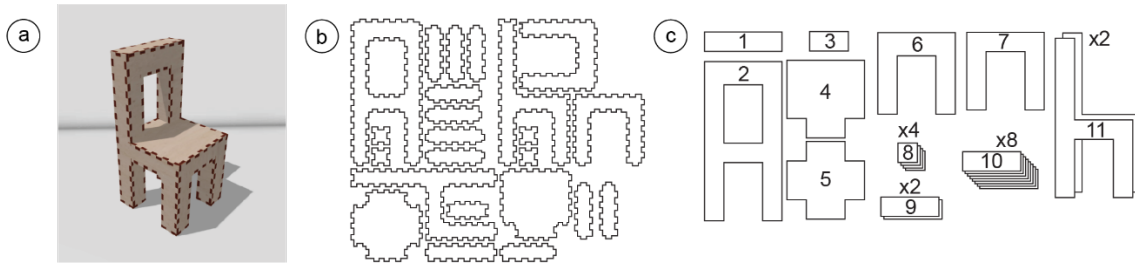


Figure 2: (a) The chair of Figure 1 consists of (b) 23 pieces a user will have to assemble into the 3D model.

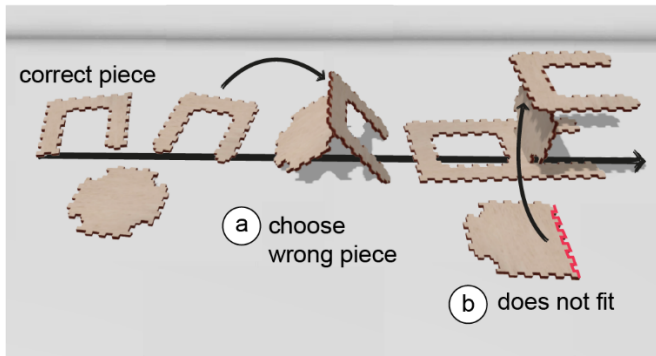


Figure 3: (a) The user has assembled an incorrect piece, but (b) does not find out the mistake until additional pieces have been assembled. We call this a propagation error.

2.1.1 FoolProofJoint makes joints different. FoolProofJoint addresses these assembly mistakes by giving each joint pair a unique finger pattern, which assures that only the correct piece can be assembled. This prevents erroneous assembly and thus also propagation errors by making users aware of their error before any piece can be assembled, giving users the opportunity to put the piece back and try again. Figure 4 shows an example. (a) The user should assemble the *front leg* piece to the *seat bottom* piece, but accidentally assembles the *backrest* piece, which just happens to have a similar envelope. (b) FoolProofJoint resolves this by making the two joint patterns distinct, preventing the incorrect piece from being assembled.

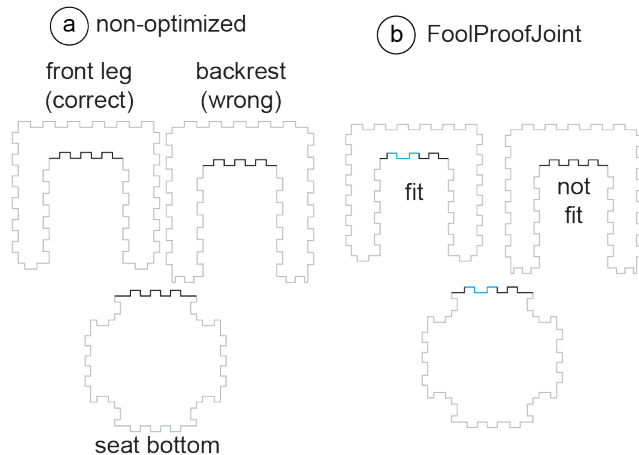


Figure 4: (a) The user wrongly assembles the backrest piece. (b) FoolProofJoint prevents this by making joint patterns distinct.

2.1.2 FoolProofJoint makes pieces interchangeable, by making their joints the same. The issue discussed above is caused by pieces *initially* looking the same to the user because they are similar, but then turning out different *later* when the user notices their differences led to assembly mistakes. However, if pieces were the same *all the way*, no problem would ever arise.

Before making any joints different, FoolProofJoint tests whether it can make pieces interchangeable. FoolProofJoint achieves this by checking whether the envelopes are the same, whether any potentially present features such as cutouts and engravings are the same, and whether the model geometry offers sufficient degrees of freedom to adjust all involved joints. If that is the case, FoolProofJoint make *the joints* of all pieces which mutually meet these criteria (we call a *piece group*) identical.

Figure 5 shows an example. Here FoolProofJoint makes the joints of all eight leg pieces of the chair model identical, thus the pieces are fully interchangeable. This not only eliminates the risk of assembly errors, but also vastly reduces the visual search to find correct pieces.

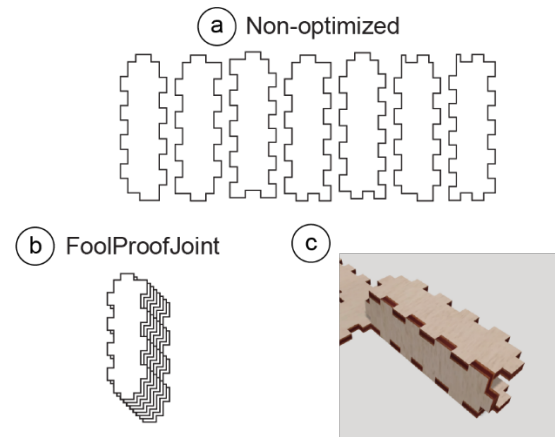


Figure 5: The chair example features eight leg pieces with the same envelope. (a) Non-optimized pieces can have different joint patterns. (b) By making their joint patterns the same, FoolProofJoint makes these pieces interchangeable, allowing users to assemble *any* of them at any fitting connection, thus reducing visual search. (c) The interchangeable pieces of legs are connected to each other by rotating 180° which is necessary to completely fill their shared edge.

2.2 Issue 2: Users accidentally re-orient symmetric pieces

The second issue that leads to assembly errors are symmetric piece envelopes. Users might try to assemble e.g., the chair's feet in any of the four possible orientations. The user may attach the piece in the correct orientation or—the user may try an incorrect orientation, causing the piece either not to fit, or to fit despite the incorrect orientation as illustrated by Figure 6, causing propagation errors.

2.2.1 Solution: FoolProofJoint makes symmetric joints. FoolProofJoint starts by testing whether it might be possible to make the joints of the piece symmetric in a way that matches its symmetries. FoolProofJoint achieves this by checking whether the piece is symmetric, whether no features such as cutouts or engravings are present that might break symmetries, and whether the model geometry offers sufficient degrees of freedom to adjust all involved joints. If that is the case, FoolProofJoint makes the piece's joints symmetric (Figure 7).

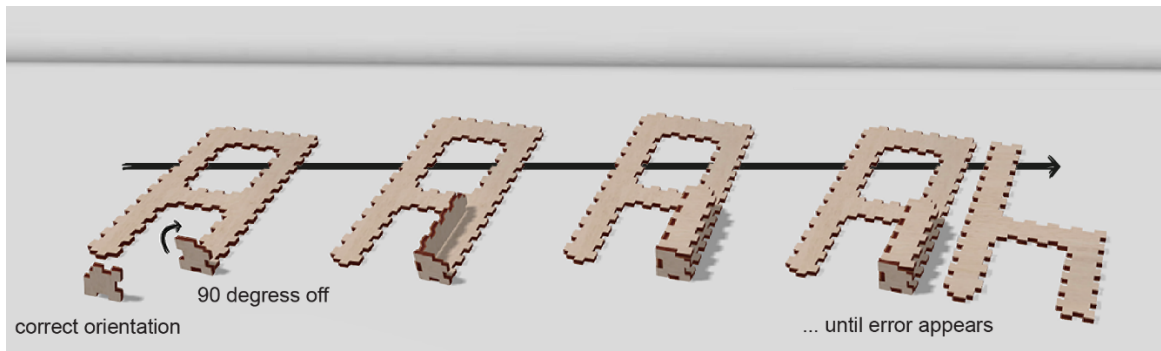


Figure 6: The user assembles a piece in the wrong orientation causing a propagation error which makes it impossible to insert the side piece a few assembly steps later.

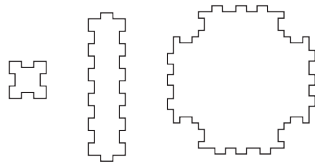


Figure 7: FoolProofJoint makes the joints of pieces symmetric to give the piece the same symmetries as its envelope.

3 ENVISIONED INTERACTION

At our demonstration, the audience will get temporary access to a website which allows them to try out the system in their web browser. The system is integrated in kyub [3], where the modified joints can be observed directly in the 3D scene or in an exported 2D cutting plan. We will be available in the virtual room to answer questions and help users if they have trouble using the software. As a fallback, we can show a video or a live demo via screencast.

4 CONCLUSION

We demonstrate FoolProofJoint, a software tool that adjusts finger joint patterns to reduce the likelihood of assembly errors by (1) giving similar pieces identical joint patterns to make them interchangeable, (2) giving pieces symmetric joint patterns according to their symmetry, and (3) giving everything else distinct joint patterns.

FoolProofJoint is as a step towards fast and most of all, reliable assembly. This step matters, as fast cutting machines and efficient design tools have sped up rapid prototyping, with assembly being the new bottleneck. FoolProofJoint also makes assembly more predictable. By eliminating the particularly time-intensive types of errors, such as propagation errors, FoolProofJoint has the potential

to help groups of users finish their assemblies roughly at the same time. This matters when running workshops and in school class, where predictable timing is of the essence.

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